

The evolution of digital wallets and cryptocurrency security

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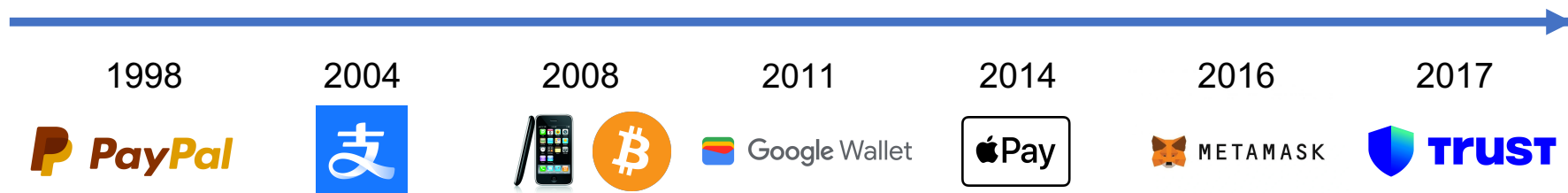
Jiawei Zhu

Part 1

The Development of Digital Wallets

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Digital wallets development timeline



Phase 1: Web Digital Wallet

Period: ~ 2000

For: Online Payment

How to pay: Card number and password

Function

- Online shopping
- Store bank cards info

Phase 2: Mobile Digital Wallet

Start: 2008 ~ now

For: Mobile Internet

How to pay: Tap/QR code/Biometric authentication

Function

- Offline shopping
- Individual transfer
- Pay the bills
- Finance and wealth

Phase 3: Crypto Digital Wallet

Period: 2015~ now

For: Blockchain and cryptocurrency

How to pay: QR code/Biometric authentication/Address

Function:

- Store addresses and keys
- Transfer without banks
- Store and send NFT
- Vote or connect to web3 apps

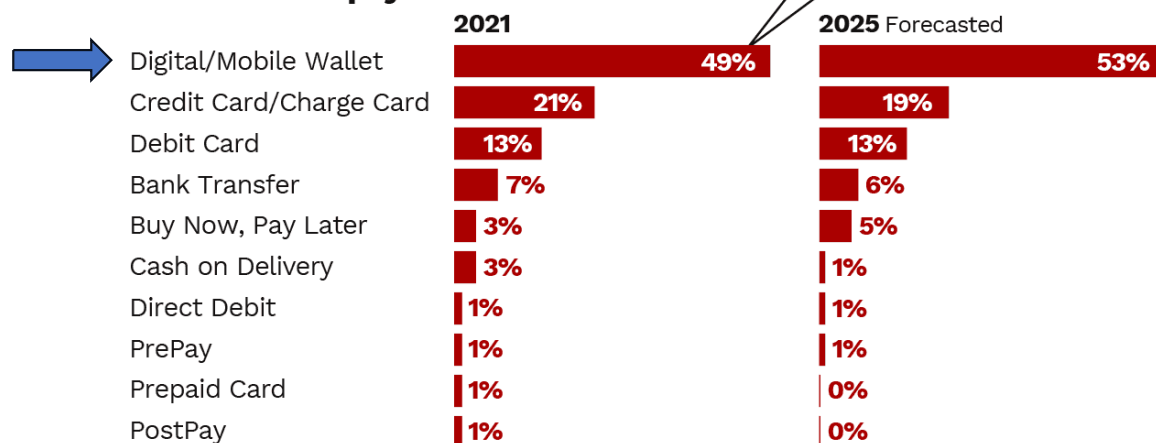
Digital wallets in payment

Fun facts about digital wallet payments :

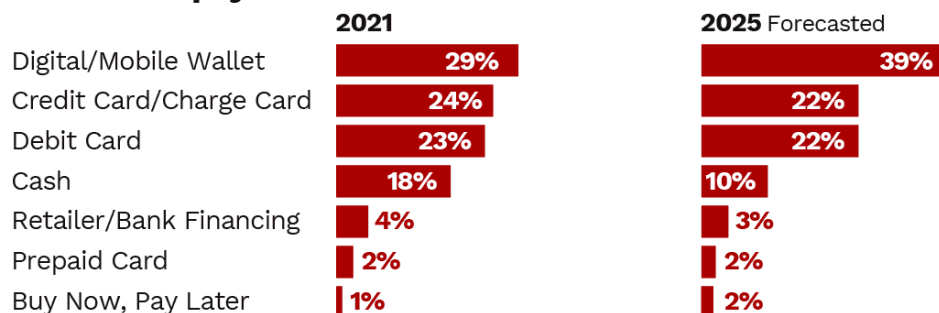
- ❓ The younger generations prefer to use electronic wallets for payment, both online and offline.
- ❓ Some developing countries have their own mainstream digital wallets.
 - China: Alipay & WeChatPay
 - India: Paytm
 - Brazil: PicPay

Where ecommerce and POS payment trends are headed worldwide

Global e-com payment methods



Global POS payment methods



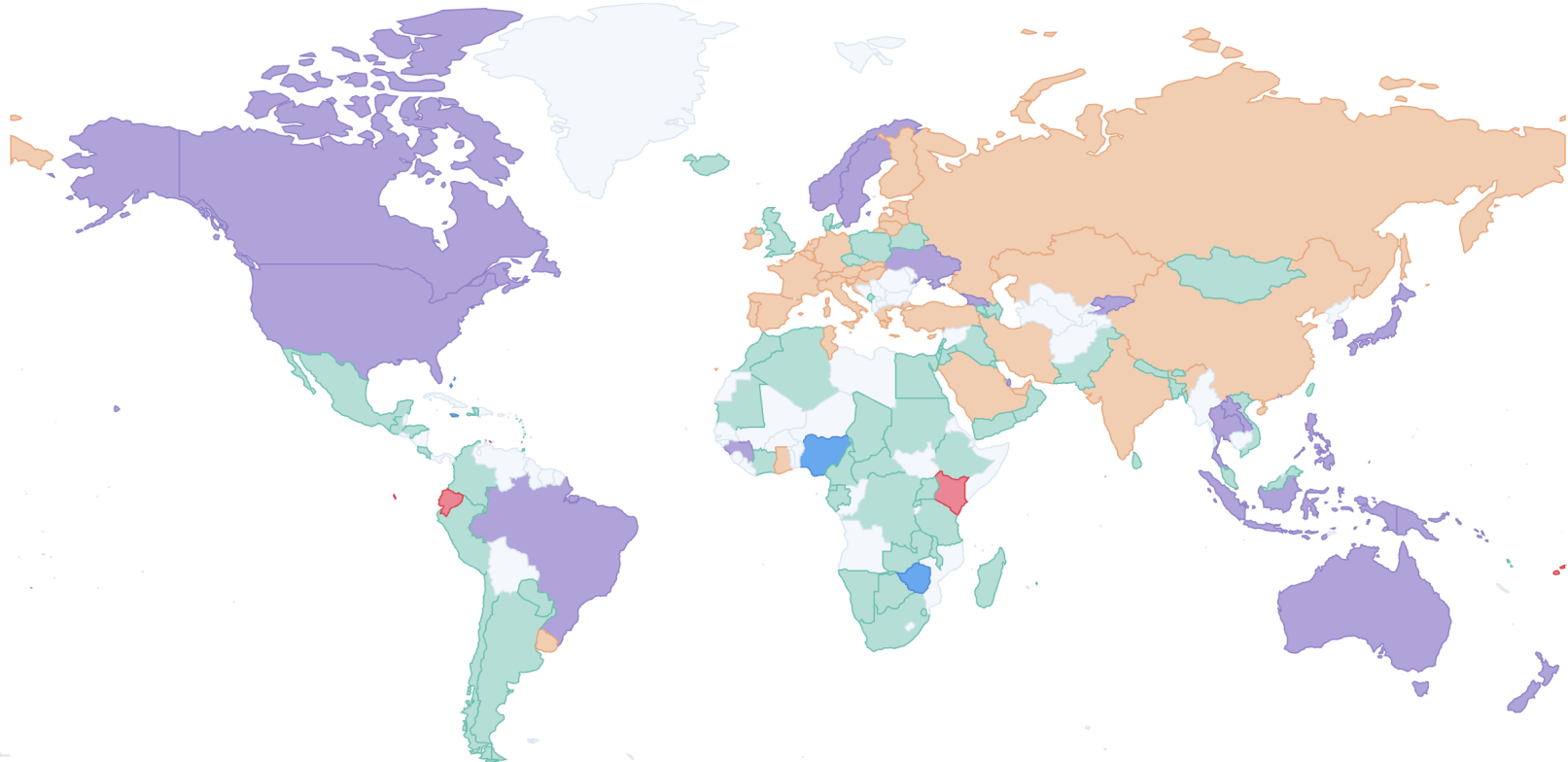
The nearest future of digital wallets: CDBC

Today's Central Bank Digital Currencies Status

Database update: April 2025 • News update: Jan, 14 25 • Total countries/regions with CBDC Projects: 115

Show filters ▾

91 Research 27 Proof of concept 24 Pilot 4 Launched 9 Cancelled 0 Inactive



Part 2

Historical Development of Cryptocurrencies and Exchanges

Member:
Wenqing Ji

2.1 The development of bitcoin

1. Bitcoin launched in 2009 by Satoshi Nakamoto – vision of trustless, peer-to-peer digital cash.
2. Early years: niche community, little value (10,000 BTC for two pizzas in 2010).





2.2 Key Milestones in Crypto Evolution

- **2013 Boom:** Bitcoin price surges from ~\$150 to >\$1,000 in two months, first major bubble (and crash) – puts Bitcoin on the map.
- **Altcoin Explosion:** Many new cryptocurrencies emerge; >1,400 cryptos launched by 2017. Ethereum (2015) introduces smart contracts, expanding use-cases.
- **Going Mainstream (2017):** Crypto market cap tops \$100 billion; increased media coverage and investor interest worldwide.

The history of Bitcoin price



BITCOIN MARKET — INFOGRAPHIC —



2.3 Rise of Cryptocurrency Exchanges

- Early trading was ad hoc; then exchanges formed to facilitate liquidity.
- Mt. Gox (Tokyo, 2010) becomes dominant – by 2012 handled ~80% of Bitcoin trades.
- Mt. Gox collapse (Feb 2014): \$450 million) to theft → trust shaken, need for security.
- Post-2014: Other exchanges rise (Bitstamp, BTC China, Kraken, etc.).
- Binance (founded 2017) rapidly grows to world's largest exchange by volume; Coinbase leads in US – exchanges now global and big business.

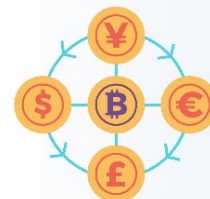
表：各衍生品交易所平台对比

	币安/Binance Centralized	火币/Huobi Global Centralized	OKEx Centralized
名称	Binance (Futures)	Huobi Futures	OKEx (Futures)
成立时间 Setup time	2019	2018	2019
交易对 (个) Number of transactions	164	214	1362
24小时交易量 (亿美元) 24-hour trading volume (billion US dollars)	1199	587	35
最活跃的交易货币组合 Most actively traded currency pairs	BTC/USDT	ETH/USDT	ETH/USDT

资料来源: coingecko, 天风证券研究所

<https://www.freepik.com/free-photos-vectors/cryptocurrency-exchange-infographic>

BITCOIN INFOGRAPHIC ELEMENTS



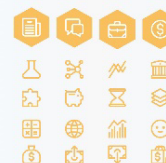
INVESTMENT



LOREM IPSUM HEADLINE



FINANCE ICONS



TRADING

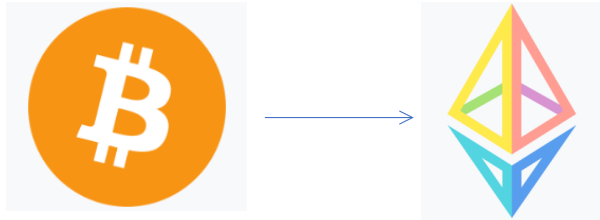


Part 3

Major Security Incidents

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- Launched in 2010 and handling over 70% of all bitcoin transactions worldwide by early 2014.
- Ceased operations abruptly.
- Hundreds of thousands of BTC, worth hundreds of millions USD, were stolen.



- Prominent cross-chain DeFi protocol.
- Exploited Aug 10, 2021 via smart-contract validation flaw.
- \$610 M siphoned; returned in 15 days.
- One of DeFi's largest security incidents by mark-to-market value.

- Launched in February 2021 as the primary sidechain for Axie Infinity, enabling high-volume, low-fee Ethereum and USDC transactions by early 2022.
- Mar 2022 Lazarus Group key breach.
- \$615 M ETH/USDC stolen.



Common Points

- **Weak hot-wallet & key management**

Many incidents (e.g., Mt. Gox, Ronin) stemmed from exposed or mismanaged private keys, enabling rapid fund theft in hot wallets without cold-storage safeguards or multi-signature controls.

- **Smart-contract flaws**

Exploits capitalized on code vulnerabilities or inadequate cross-chain validation logic, demonstrating gaps in secure contract design and audit processes.

- **Centralized single points of failure**

Both centralized exchanges and so-called “decentralized” bridges relied on a small set of keys, nodes, or administrators—once compromised, the entire system collapsed.

- **Insufficient monitoring & response**

Major hacks often went undetected for days, revealing a lack of real-time on-chain monitoring, alerting, and rapid incident-response procedures.

Ecosystem Challenges

- **Immature security practices**

Absence of industry standards for private-key custody, multi-sig policies, and cold-storage protocols leaves critical assets exposed.

- **Inadequate contract audit frameworks**

Lack of unified vulnerability-disclosure channels and formal certification bodies means code flaws frequently go unnoticed until exploitation.

- **“Decentralization” vs. operational centralization**

Despite on-paper decentralization, real-world implementations hinge on a few trusted nodes or keys, creating hidden centralization risks.

- **Regulatory & legal gaps**

No clear legal frameworks or investor-protection mechanisms slow down post-breach recovery and increase remediation costs.

Part 4

Security standards and protection strategies of Cryptocurrencies

Member:
heqing Shi

4.1 Security standards



CCSS (CryptoCurrency Security Standards)

- **Publication date:** 2014
- **Organization:** C4 (CryptoCurrency Certification Consortium)
- An industry standard specifically designed. including key generation, key storage and access control, key backup and recovery-to-backup, and audit Logs.
- **Application cases:** *BitGo* adopts the CCSS standard to secure the private key management, access control and audit process.

FIPS 140-2/3 (Federal Information Processing Standard 140-2/3)

- **Publication date:** 2001/2023
- **Organization:** NIST(National Institute of Standards and Technology)
- It is a globally recognized security certification standard for encryption modules, which is widely used in hardware wallets and cold wallets.
- **Application cases:** The hardware wallet *Ledger Nano X* uses a FIPS 140-2 certified security chip to ensure that the user's private key is stored in compliance with government encryption standards.



ISO/TC 307 (Blockchain and Distributed Ledger Technologies)

- **Publication date:** 2018
- **Organization:** ISO(International Organization for Standardization)
- Regulate and standardize the consensus mechanism and smart contract , and also proposed requirements for identity management and data privacy, to ensure the safe landing of blockchain applications
- **Application cases:** **The Australian government** adopted the ISO/TC 307 standard to regulate identity authentication and smart contracts when developing a national blockchain commodity traceability platform.



4.2 protection strategies of Cryptocurrencies

1. Wallets and key management

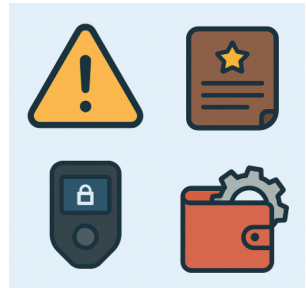
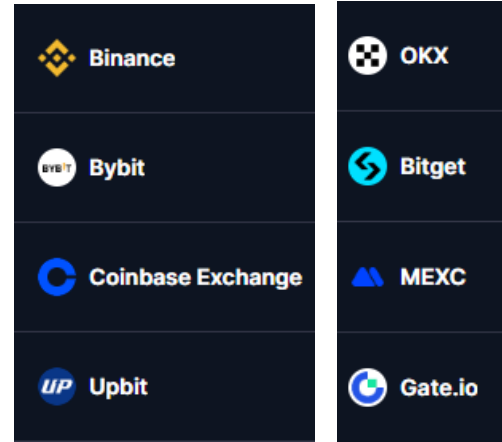
- **Separation of hot and cold wallets** : Hot are used for daily transactions, and cold are used for offline long-term storage. (Example: **Ledger / Trezor**-hardware wallets.)
- **Multi-signature**: Multiple people or devices sign a transaction at the same time, reducing the risk of a single point of disclosure.
- **Multi-party Computation (MPC) wallet**: Split the private key and store it in multiple nodes, and then conduct joint signature.

2. Exchange platform security protection

- **Real-time risk control system** :By monitoring the abnormal IP to prevent abnormal transactions in time. such as: short time high-frequency operation / large transfer.
- **Two-factor authentication (2FA)**:User login, withdrawal, and other operations require a second verification, such as SMS verification code.
- **Bug Bounty**: Rewarding white-hat hackers for early detection of potential vulnerabilities.

3. User security awareness

- Prevent phishing attacks / Keep mnemonic phrases safe / Use hardware wallets / Regularly update wallet software.



Part 5

Summary

Member:
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5.1 Where Wallets Are Heading: Emerging Security Trends

AI-driven fraud detection

- Automatically detects abnormal transactions and device behavior
- Enhances real-time anti-fraud capabilities

Biometric and behavioral authentication

- Uses fingerprint, facial recognition, and behavioral data
- E.g., typing rhythm, gait analysis for identity verification

Zero-knowledge proof & MPC (Multi-Party Computation)

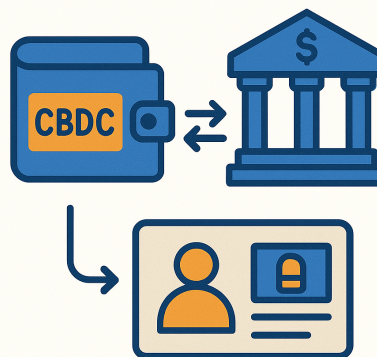
- Enables authentication without revealing private information
- Distributed key signing (MPC) is being adopted in commercial wallets

Post-quantum cryptography

- Resilient encryption algorithms against quantum computing threats
- Prepares wallets for next-generation cybersecurity challenges

Integration with CBDC and National ID

- Wallets will integrate with Central Bank Digital Currencies (CBDCs)
- Expected to become the carrier of national digital ID systems



5.2 Security is Evolving — But Not Solved Yet

Remaining Challenges:

User-side vulnerability Mnemonic loss, phishing, and human errors are still the main causes of breaches.

Interoperability & standard gaps Multiple wallets and chains lack unified security protocols and standards.

Centralization in "decentralized" systems Some bridges and MPC setups rely on few nodes, posing single-point-of-failure risks.

Lagging regulations Lack of clear legal frameworks for wallet security and post-incident accountability in most countries.

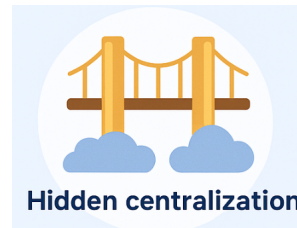
Security–usability trade-off The more secure the wallet, the harder it becomes for the average user to manage.

Summary:

From Web → Mobile → Crypto, wallets have grown in functionality — and attack surface.

Standards (like CCSS, FIPS, ISO) are evolving, but need better user alignment and regulation.

The future of wallet security is more than "technical upgrade" — it requires integration of identity, data, behavior, and governance.



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Part 6

Let's Test!

<https://www.mentimeter.com/app/presentation/al2ntp2y1e6xagerpqe7mbqaymk33pdj/edit?question=dtjvb6jxoz93>

